

NGUYEN GIA HUY

Python Backend Engineer

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PROFESSIONAL SUMMARY

Python Backend Engineer working on core banking at Vikki Digital Bank, developing the Lending and Overdraft products as Python Smart Contracts on Thought Machine Vault Core. Previously built ETL pipelines on AWS for a large-scale utility platform. Most of the work involves financial logic, data processing, and backend integrations.

TECHNICAL SKILLS

Languages:	Python, SQL, Golang
Backend:	FastAPI, REST APIs, gRPC
Databases & Messaging:	PostgreSQL, Redis, Kafka
Cloud:	AWS (Lambda, ECS, RDS, S3, Athena), Docker
Core Banking:	Thought Machine Vault Core

WORK EXPERIENCE

Python Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Developed and maintained the Lending and Overdraft products as Python Smart Contracts on Thought Machine Vault Core.
- Implemented interest accrual for both products using Vault Core's scheduled events, along with penalty and late-fee calculation for overdue payments.
- Implemented repayment and settlement logic, which directly updates customer balances on the core ledger.
- Integrated Vault Core with external backend services used in core banking operations and transaction workflows.
- Rebuilt the simulation testing framework used to test the products, moving it from Gauge to Golang and cutting test runs from **24 minutes to just over 50 seconds**.
- Built a Golang account migration system (ECS, Lambda, RDS) that brought migration time down to **27 minutes from 2.5 hours**.
- Deployed a new Vault Core version on AWS (EKS, RDS, ECR, ELB) with a CI/CD pipeline for build, test, and deployment.

Software Engineer – Data Processing 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Worked on the data processing side of au Denki's platform: ETL pipelines on AWS Lambda and ECS that transform large-scale utility data, with results stored in S3, Athena, and RDS for analytics and operations.
- Increased the throughput of the batch jobs by **150%** through parallel execution, a pipeline redesign, and leaner data access.
- Rebuilt the ingestion and indexing flows for the platform's move from Elasticsearch to OpenSearch, improving stability, consistency, and query speed.
- Managed infrastructure with Terraform and AWS SAM and set up monitoring in CloudWatch and DataDog.

Software Engineer 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Implemented tenant-isolated authentication and authorization for **20+** customer organizations using Keycloak and RBAC.
- Ran Kong Gateway for **100+** microservices, including their service-level and network-level access control.
- Operated **50+** production Kubernetes clusters on VNPAY's OpenStack cloud; provisioning, scaling, and zero-downtime upgrades were automated with Cluster API.
- Automated about **80%** of data center provisioning and infrastructure configuration.

Blockchain Developer

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Processed on-chain events from **5+** chains in event-driven services and stored the results in off-chain databases and APIs, with consistent transaction handling.
- Developed and reviewed **15+** smart contracts for DeFi and NFT use cases in Solidity and Move.

SELECTED PROJECTS

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on LiveKit, integrating OpenAI models for live transcription, voice translation, and meeting summaries.
- Designed the real-time translation pipeline and improved how conversation context is handled, which made translations more accurate and coherent.

DattingQ | Software Engineer

Dating platform

- Built a backend on gRPC-Gateway, Kafka, MongoDB, and Redis that handled up to **1.5K** requests per second at 80–100 ms latency, with multi-cluster failover and **99.95%** uptime.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. in Information Technology, Knowledge Engineering

DevOps on AWS Specialization | Coursera